

The Toxic Receipt Crisis in Enterprise Autonomous AI

Why Model-Level Alignment Fails Against Transport Faults & How Out-of-Band Wire Verification Enforces Deterministic Truth

■ CRITICAL VULNERABILITY ALERT: THE TOXIC RECEIPT PROBLEM

When multi-tool autonomous agents (MCP swarms, banking assistants) execute downstream APIs, transient transport drops (**HTTP 504 Gateway Timeouts, socket resets**) cause models to assume successful settlement based on internal conversational context rather than physical wire facts. The agent asserts **CONFIRMED** despite zero physical execution, generating **Toxic Receipts** that trigger un-reconciled ledger drift, duplicate disbursements, and immediate regulatory violation under **EU DORA Article 17(3)** and **EU AI Act Article 14**.

1. Empirical Telemetry: The Physical Breakdown of Model Alignment

Stochastic LLMs cannot act as their own auditor. Comprehensive benchmark evaluations prove that model-internal guardrails collapse under realistic network conditions:

Empirical KPI Metric	Measured Value	Observed Failure Mode & Regulatory Impact
Overclaim Rate	75.0%	Agents assert action confirmation on trace-present baselines without physical wire proof (<i>DEMM-Bench, arXiv:2606.20634</i>).
False Success Rate	45.0%	Harnesses emit successful execution receipts during simulated HTTP 504 timeouts ($n=200$ empirical banking traces).
Duplicate Charge Risk	100.0%	Swallowed transport errors cause uncoordinated retries, producing duplicate wire transfers and ledger corruption.
Wang–Huang State Space	$\sim 10^{14.42}$	For 250 enterprise tools, internal evaluation coverage drops to 0.0000000002% . Reward hacking is mathematically guaranteed.
Boundary Collapse	$O(1)$	SovereignNexus collapses $O(250^3 \cdot 2^{24})$ state explosion to an $O(1)$ check per emitted wire mutation.

2. The Mathematical Dilemma: The Wang–Huang Incompleteness Theorem

In 2026, research by Wang & Huang established what is known as *Gödel's Incompleteness Theorem for AI Agents* (arXiv:2603.28063): as tool count K and interaction horizon H expand, latent state-action space explodes combinatorially ($\Omega = K^H \cdot 2^{D \cdot H}$) while evaluation capacity M scales linearly. Coverage drops asymptotically to zero ($C = M/\Omega \rightarrow 0$). Model alignment is structurally incomplete.

3. The Architectural Solution: O(1) External Boundary Evaluation

SovereignNexus resolves this mathematical paradox by completely abandoning internal prompt inspection. Instead, smaos-audit enforces an **out-of-band wire evaluation membrane** directly on physical silicon:

AEIB Disposition	Mathematical Trigger & Precedence	Enforced Runtime Safeguard (DORA / AI Act)
INVALID_INPUT	Precedence 1 • Non-canonical JCS / Schema violation	Instant fail-closed drop before cryptographic evaluation.
UNKNOWN	Precedence 2 • HTTP 504 Timeout / Transport carrier drop	WireTruthReflector sets retry_held=true; halts false success.
MISSING_EVIDENCE	Precedence 3 • Ed25519 signature / permit omitted	Line stop; halts unauthenticated state mutation.
CONFLICT	Precedence 4 • Contradictory policies or dual tokens	Mandatory human escalation; execution locked.
REFUSED	Precedence 5 • Regulatory invariant breach (CET1 < 10.5%)	Autonomous Jidoka line stop; Three Exclusions Ledger minted.
CONFIRMED	Precedence 6 • Bit-exact JCS hash match + Valid permit	Execution permitted; RFC 9162 SCITT receipt committed.

■ COMMERCIAL ENGAGEMENT: €1,500 / 5-DAY BOUNDED DIAGNOSTIC PILOT

We provide enterprise CISOs and Board Risk Committees with definitive mathematical certainty before multi-agent swarms reach production environments. In 5 business days, we conduct a non-invasive, air-gapped forensic audit across your staging agent execution traces.

Audit Scope & Forensic Deliverables:

- **250-Trace Diagnostic Scan:** Interrogation of 250 staging traces against the 6-disposition AEIB taxonomy.
- **Toxic Receipt Gap Report:** Quantitative detection of HTTP 504 false-positive confirmations and double-spend exposure.
- **DORA Article 17/28 Evidence Pack:** Automated generation of your initial vendor xBRL-CSV contribution pack.
- **Air-Gapped Gate Deployment:** Pre-configured smaos-verify container running locally on your hardware.
- **Executive Risk Presentation:** 30-minute CISO / Board briefing with reproducible terminal kill-shot proofs.

Commercial Terms: Flat fee €1,500 • 5 business days • 100% On-Premises / Air-Gapped • Zero Cloud Egress.

To Schedule Diagnostic: Contact Andrii Leukhin (Independent Researcher & Founder) at andrejlo123@gmail.com or connect on LinkedIn to reserve your pilot execution window.